SIGMIDI FOOT CONTROLLER

v0.1 / KICAD ELECTRICAL SCHEMATIC / external board pins only

Continuous wires connect every fitted terminal. Dots = junctions; crossings without dots = no connection. Component placement is schematic.

Nano and LCD pins keep physical order. Nano: JP1 left 1–15, JP2 right 15–1, USB-C up. Outlines are schematic.

3V3 / Nano logic supply

+5V LED

J1 5.5 x 2.1mm
9V DC / center-negative
SLEEVE (+)
CENTER (–)

C1
47uF / 35V

U3 S13V10F5
Pololu #4083 / 1178565
VIN
GND
5V OUT

USB-C / MIDI

U1 Arduino Nano ESP32
Component-side top view / USB-C up

JP1:1 D13
JP1:2 / 3V3 3V3
JP1:3 B0
JP1:4 / GPIO1 A0
JP1:5 / GPIO2 A1
JP1:6 A2
JP1:7 / GPIO4 A3
JP1:8 / GPIO11 A4
JP1:9 A5
JP1:10 A6
JP1:11 A7
JP1:12 / VBUS VBUS / VUSB
JP1:13 B1
JP1:14 / GND GND
JP1:15 / VIN VIN

JP2:15 / GPIO47 D12
JP2:14 / GPIO38 D11
JP2:13 / GPIO21 D10
JP2:12 / GPIO18 D9
JP2:11 / GPIO17 D8
JP2:10 / GPIO10 D7
JP2:9 / GPIO9 D6
JP2:8 / GPIO8 D5
JP2:7 / GPIO7 D4
JP2:6 / GPIO6 D3
JP2:5 / GPIO5 D2
JP2:4 / GND GND
JP2:3 RST
JP2:2 D0 / RX0
JP2:1 D1 / TX0

All printed labels retained; unused pads left unwired.

U2 Waveshare 24382
1 BL
2 RST
3 D/C
4 CS
5 SCL / CLK
6 SDA / DIN
7 GND
8 VCC
P2 / GH1.25 / SPI / pins 1–8

J2 ACJS-MHDC / passive pedals
TOP / pedal 1
11 RING
12 TIP
10 SLEEVE
BOTTOM / pedal 2
5 RING
6 TIP
4 SLEEVE
1/2/3/7/8/9: NC; pad 13: mechanical / NC

R20 5.1k
R30 5.1k

USB VBUS sense / 3.0V at 5V
R50 100k
R51 150k

Pololu 2535 / optional LEDs

LED1 / B
LED2 / A
LED3 / 2
LED4 / 1
LED5 / C
LED6 / 3
LED7 / D
LED8 / 4
LED9 / E
LED10 / 5
2535
DIN
5V
GND
DOUT
NC

R21 10k
R31 10k

SW11 ENCODER
A
20 detents + click
B
Internal pull-ups
SW1
No external R/C
C
SW2

MODULAR FOOTSWITCH LADDER

R7 10k
R8 20k
C5 100nF
R9 1k
R10 1k
R11 1k
R12 1k
R13 1k
R14 1k
R15 1k
R16 1k
R17 1k
R18 1k

SW1 B — T1 / LED1
SW2 A — T2 / LED2
SW3 2 — T3 / LED3
SW4 1 — T4 / LED4
SW5 C — T5 / LED5
SW6 3 — T6 / LED6
SW7 D — T7 / LED7
SW8 4 — T8 / LED8
SW9 E — T9 / LED9
SW10 5 — T10 / LED10

COMMON GND

PASSIVE PEDALS

Pot: outer legs RING / SLEEVE; wiper TIP.
Variable resistor: TIP to SLEEVE; RING unused.
0–10k per input. Bias HIGH in resistor mode;
high impedance in pot / disabled / boot states.

Netlist, BOM and pin references: design/electrical/. All 46 components and 41 nets checked against the connected drawing.

ASSEMBLY

Set 4 / 6 / 8 / 10 switches: calibrate installed taps.
SW11 contact pin arrangement is functional.
Native USB-C carries USB MIDI and ROM flashing.
USB-only supply does not power the external LED rail.

LED PACKAGE

Lens left, leads right, flat side down.
Top to bottom: DIN / 5V / GND / DOUT.
Only LEDs and wires; direct D7 / GPIO10 data.
Check 3.3V data acceptance on the actual LED batch.

Sheet: /
File: sigmidi-foot-controller.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.1 | Id: 1/1